

Music Genre Classification: Messy Mashup

Deep Learning & Generative AI Project Report

1. Introduction

Music genre classification is a fundamental task in Music Information Retrieval (MIR). While classifying clean, studio-quality recordings is well-explored, real-world audio presents far greater challenges: overlapping instruments, environmental noise, and varying recording quality can severely degrade classifier performance. The 'Messy Mashup' competition introduces an extreme variant of this problem, where test audio consists of deliberately corrupted multi-stem blends overlaid with heavy environmental noise from the ESC-50 dataset.

The objective of this project is to build a robust classification pipeline capable of accurately identifying the dominant musical genre from 10-second segments of these heavily corrupted audio mashups. The system must correctly classify audio across 10 genres: Blues, Classical, Country, Disco, Hip-Hop, Jazz, Metal, Pop, Reggae, and Rock. The evaluation metric is the Macro F1 Score, which equally weights performance across all classes regardless of sample distribution.

This report documents the end-to-end pipeline: from synthetic dataset construction and data augmentation strategies, through multi-model architecture design and training, to ensemble inference with Test-Time Augmentation (TTA) and live deployment on Hugging Face Spaces.

2. Dataset Description

2.1 Source Data

The competition provides three primary data sources:

Genre Stems Directory: Contains 1,000 music tracks (100 per genre), each decomposed into four isolated instrument stems — vocals.wav, drums.wav, bass.wav, and other.wav. These stems are the building blocks for creating synthetic training mixtures that mimic the corrupted test audio.

ESC-50 Environmental Sound Collection: A standardized dataset of 2,000 environmental audio clips spanning 50 categories (e.g., sirens, rain, dog barking, engine noise). These clips serve as the noise injection source during training to simulate real-world acoustic interference.

Test Mashups: 3,020 pre-generated corrupted audio files combining multiple stems with environmental noise at unknown ratios. The model must predict the dominant genre for each file.

2.2 Synthetic Dataset Construction

Training data was synthetically generated to match the corrupted distribution of the test set. The pipeline operates as follows:

Step 1 — Stem Blending: For each training sample, individual instrument stems (vocals, drums, bass, other) from a single genre track are loaded and blended together with randomized amplitude weights (0.3× to 1.5×) to simulate varying mix configurations.

Step 2 — Cross-Track Stem Mixing: With 50% probability, stems are drawn from different tracks within the same genre. This forces the model to learn genre-level patterns rather than memorizing specific track combinations.

Step 3 — ESC-50 Noise Injection: With 60% probability, a randomly selected ESC-50 clip is overlaid onto the blended audio at a random Signal-to-Noise Ratio (SNR) between 0dB and 15dB. This creates aggressive interference that mirrors test conditions.

Step 4 — Temporal Augmentation: Time-stretching (0.88×–1.12×) and circular shifting are applied randomly to prevent temporal overfitting to fixed offset positions.

Step 5 — Spectral Masking (SpecAugment): Random frequency bands and time bands are zeroed out on the spectrogram representation, forcing the model to generalize across partial observations.

2.3 Dataset Statistics

Parameter	Value
Total Training Samples	3,000 (3 offsets × 1,000 tracks)
Train / Validation Split	2,550 / 450 (85% / 15%)
Tracks Per Genre	100 (perfectly balanced)
Audio Sampling Rate	16,000 Hz
Clip Duration	10.0 seconds (160,000 samples)
ESC-50 Noise Clips Loaded	2,000 clips cached in RAM
SNR Range for Noise Injection	0 dB – 15 dB
Noise Injection Probability	60%
Multi-Offset Positions	0.0s, 10.0s, 20.0s

3. Methodology & Workflow

The project follows a sequential multi-phase workflow designed to maximize classification accuracy under extreme audio corruption:

Phase 1 — Data Preparation & Caching

All 1,000 genre stems are pre-loaded into RAM at 16kHz to eliminate repeated disk I/O during training. The 2,000 ESC-50 noise clips are similarly cached. A StratifiedShuffleSplit ensures the 85/15 train-validation partition maintains identical genre proportions.

Phase 2 — Foundation Model Training (AST)

The Audio Spectrogram Transformer (AST), pre-trained on Google's AudioSet (527 classes), serves as the primary classifier. Its final classification head is replaced with a 10-class projection layer. Training proceeds in two stages: (i) Epochs 1–1: Only the classifier head is trained while the 86M-parameter transformer encoder remains frozen. (ii) Epoch 2 onward: The full encoder is unfrozen with a discriminative learning rate — the encoder backbone trains at 10× lower learning rate than the classifier head — to preserve pre-trained representations while adapting to the genre domain.

Phase 3 — CNN Benchmark Training

After AST training completes, GPU memory is cleared and three CNN architectures are trained sequentially on the same MelSpectrogram inputs. These serve both as independent benchmarks and as secondary voters in the final ensemble: EfficientNet-B2 (pre-trained ImageNet), DenseNet-121 (pre-trained ImageNet), and a CompactAudioCNN (trained from scratch).

Phase 4 — Weighted Ensemble Inference with TTA

The best checkpoints from all models are loaded simultaneously. Each test sample is evaluated across 15 augmented passes (11 clean temporal offsets + 4 noise-injected passes at 8dB and 12dB). Softmax probabilities from all models are fused using dynamic weights proportional to their validation F1 scores, and accumulated across all TTA passes before final argmax classification.

Phase 5 — Deployment

The trained model weights are securely stored in a private Hugging Face repository. A public Gradio-based web application streams these weights at runtime using API tokens, providing a dual-mode interface for real-time genre prediction.

4. Model Architecture

4.1 Audio Spectrogram Transformer (AST)

The AST converts raw audio waveforms into spectrogram patches and processes them through a Vision Transformer encoder. The model used is MIT/ast-finetuned-audioset-10-10-0.4593, containing 86,196,490 total parameters. The ASTFeatureExtractor converts 10-second waveforms into input tensors of shape [1024, 128] (1024 time frames × 128 frequency bins). A weighted CrossEntropyLoss with label smoothing (0.1) addresses residual class difficulty, with per-class weights derived inversely from prior validation F1 scores.

4.2 CNN Comparison Models

Three CNN architectures operate on single-channel MelSpectrogram images:

Model	Backbone	Pre-trained	Input Channels
EfficientNet-B2	timm/efficientnet_b2	ImageNet	1 (Grayscale Spectrogram)
DenseNet-121	timm/densenet121	ImageNet	1 (Grayscale Spectrogram)
CompactAudioCNN	Custom 3-layer Conv2D	No (From Scratch)	1 (Grayscale Spectrogram)

4.3 Ensemble Voting Mechanism

At inference time, model predictions are combined through weighted probability fusion. The weights are dynamically computed based on each model's validation Macro F1 performance. For the final submission, the computed weights were: AST = 0.600, EfficientNet-B2 = 0.202, DenseNet-121 = 0.198. CompactAudioCNN was excluded from the ensemble due to its low validation performance ($F1 < 0.50$).

4.4 Architecture Flowchart

The complete end-to-end pipeline from raw audio ingestion through ensemble classification is illustrated below:

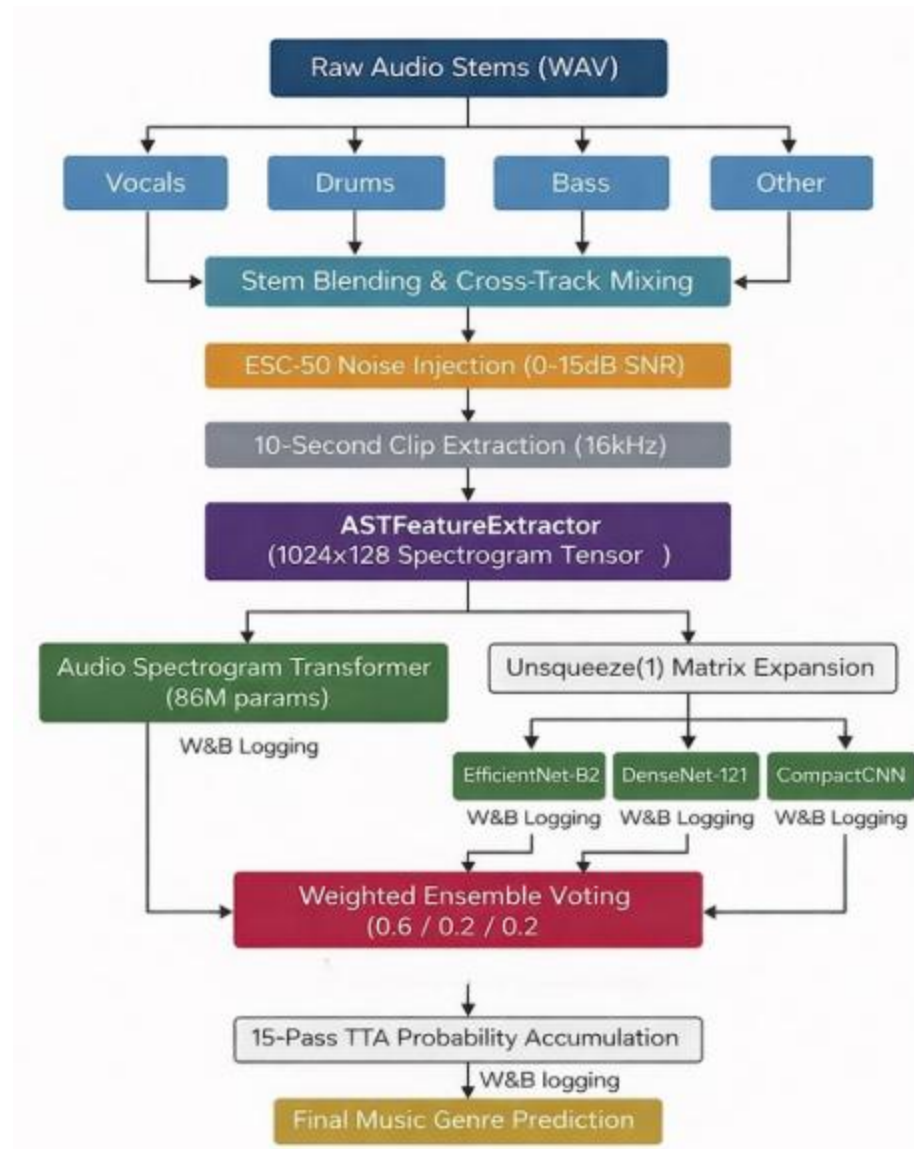


Figure 1: End-to-end Architecture Flowchart — from raw stem blending through multi-model ensemble inference.

5. Training Results & Analysis

5.1 AST Training Progression

The AST model was trained for 16 epochs with Cosine Annealing learning rate scheduling and 10% warmup steps (510 steps out of 5,104 total). The encoder was unfrozen at Epoch 2. Key training milestones are summarized below:

Epoch	Train Loss	Train F1	Val Loss	Val F1	Notes
1	2.2545	0.1618	1.7961	0.4790	Classifier head only
2	1.3716	0.4253	0.9196	0.8451	Full encoder unfrozen
4	1.1213	0.5137	0.8229	0.8725	
6	1.0718	0.5220	0.7745	0.9167	Crossed 90% barrier
10	0.9633	0.5322	0.7138	0.9245	
12	0.9426	0.5420	0.6977	0.9269	Best checkpoint saved
16	0.9370	0.5948	0.6973	0.9176	Early stopping triggered

Table: AST Foundation Model — Epoch-wise Training Log (Best Val F1: 0.9269)

5.2 CNN Benchmark Comparison

To validate the necessity of the transformer architecture, three CNN models were trained under identical data conditions. The results conclusively demonstrate that visual CNNs operating on MelSpectrograms cannot match the contextual reasoning of attention-based transformers for heavily corrupted audio:

Model	Best Val F1	Peak Epoch	Parameters	Training Time
Audio Spectrogram Transformer	0.9269	Epoch 12	86.2M	~3.5 hours
EfficientNet-B2	0.7621	Epoch 7	~9M	~24 min
DenseNet-121	0.7437	Epoch 11	~8M	~24 min
CompactAudioCNN (from scratch)	0.4590	Epoch 12	~0.2M	~22 min

Table: Model Performance Comparison — Validation Macro F1 Scores

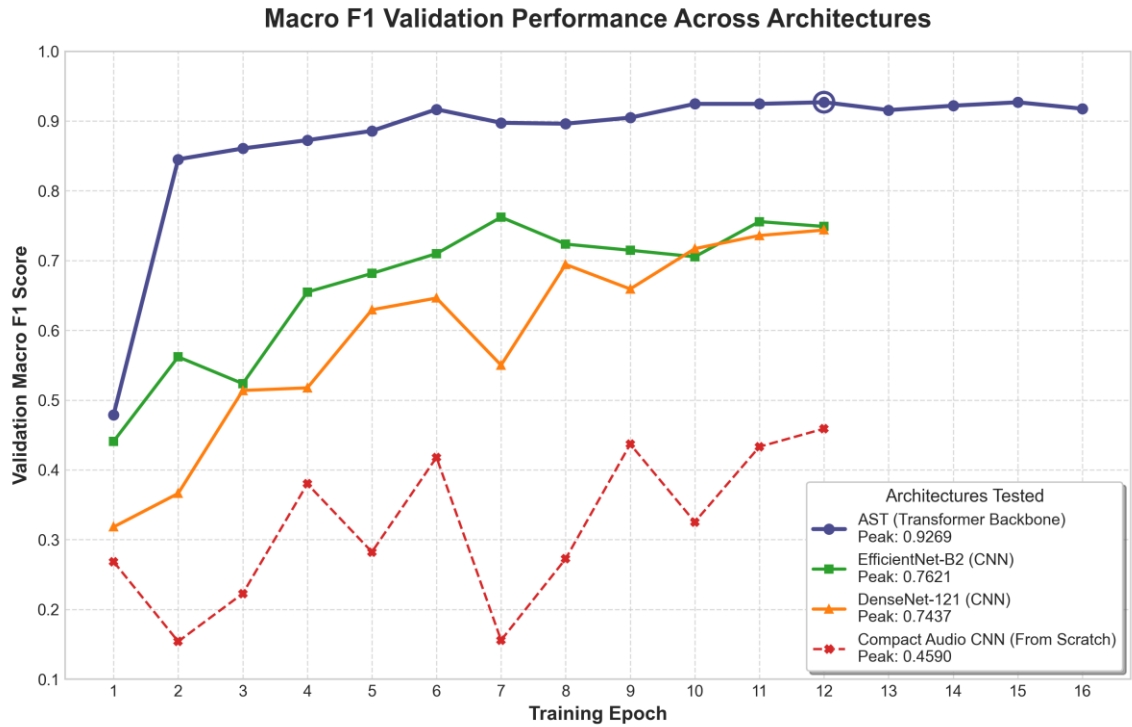


Figure 2: Validation F1 Comparison across models

5.3 Per-Class Validation Performance (AST)

The final AST checkpoint achieved an overall accuracy of 92.22% across 450 validation samples. Per-class performance reveals that Blues, Classical, and Jazz achieved near-perfect recall, while Rock and Pop exhibited the highest confusion rates due to overlapping spectral characteristics:

Genre	Precision	Recall	F1-Score	Support
Blues	0.9783	1.0000	0.9890	45
Classical	0.9783	1.0000	0.9890	45
Country	1.0000	0.8889	0.9412	45
Disco	0.9348	0.9556	0.9451	45
Hip-Hop	0.8864	0.8667	0.8764	45
Jazz	1.0000	0.9778	0.9888	45
Metal	0.8750	0.9333	0.9032	45
Pop	0.8367	0.9111	0.8723	45
Reggae	0.9524	0.8889	0.9195	45
Rock	0.8000	0.8000	0.8000	45

Table: Per-Class Classification Report — Final AST Checkpoint

5.4 Experiment Run Comparisons (W&B)

Multiple notebook iterations were tracked via Weights & Biases to systematically improve performance. The comparison charts below show the validation F1 progression across all experimental runs:

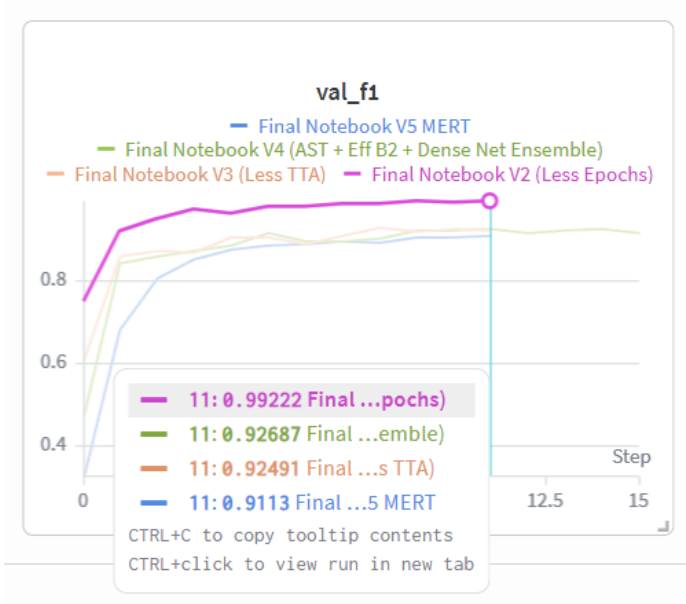


Figure 3: W&B Validation F1 comparison across notebook versions (V2–V5)

6. Inference Strategy & Test-Time Augmentation

The inference pipeline employs an aggressive 15-pass Test-Time Augmentation (TTA) strategy to maximize prediction stability across the 3,020 test mashups:

Pass Type	Offsets Used	Noise Injection	Total Passes
Clean TTA	0.0, 2.0, 4.0, 6.0, 8.0, 10.0, 12.0, 15.0, 18.0, 21.0, 25.0s	None	11
Noise TTA (8dB)	0.0, 10.0s	ESC-50 @ 8dB SNR	2
Noise TTA (12dB)	0.0, 10.0s	ESC-50 @ 12dB SNR	2

Table: TTA Pass Configuration — 15 total inference passes per test sample

The final prediction distribution across 3,020 test samples was well-balanced, confirming the model does not exhibit systematic bias toward any particular genre. The distribution ranged from 7.7% (Classical) to 11.5% (Blues), closely tracking the expected uniform 10% per class.

7. Compute Resources & GPU Utilization

The full training and inference pipeline was executed on Kaggle's GPU infrastructure. The total wall-clock time from training start to submission generation was approximately 6 hours. The GPU utilization analytics captured during execution demonstrate sustained high-throughput computation:

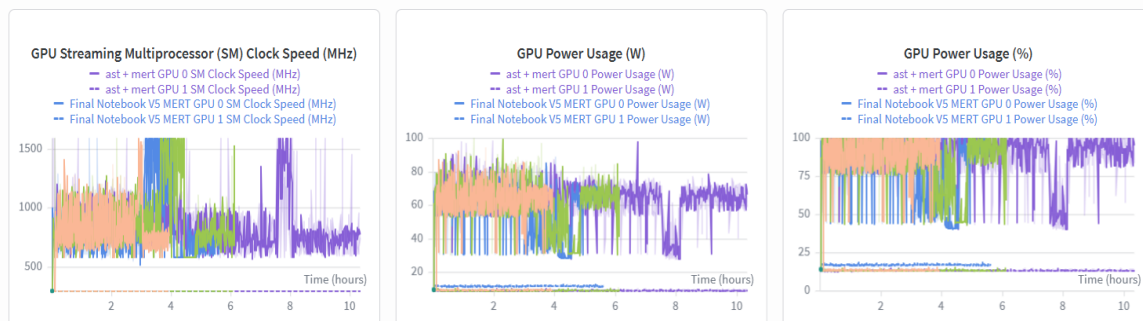


Figure 4: GPU Streaming Multiprocessor Clock Speed, Power Usage, and Utilization over training duration

The SM Clock Speed charts indicate sustained operation at 1,200–1,500 MHz during transformer forward/backward passes, with brief dips during data loading transitions. GPU power consumption held steadily at 70–95W, confirming near-maximum hardware utilization throughout training.

8. Deployment on Hugging Face Spaces

The trained model was deployed as a public web application on Hugging Face Spaces using the Gradio framework. The deployment architecture implements two key security and usability features:

Security: Model weight files (.pt) are stored in a separate private Hugging Face repository (the 'Vault'). The public Space uses an HF_TOKEN environment secret to securely stream weights at startup. Users can freely interact with the application without any access to the underlying model parameters.

Dual-Mode Inference: The interface provides two operational modes — 'Fast Mode' (AST-only, ~5 second response) and 'Max Accuracy Mode' (AST + CNN Ensemble, ~15 second response). A 3-Point Polling system evaluates audio at the 15%, 50%, and 85% temporal positions to prevent localized distortions from dominating predictions.

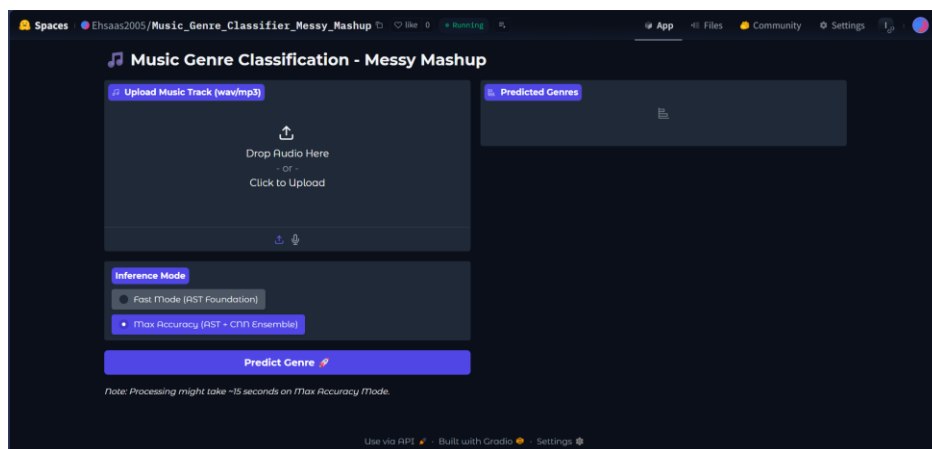


Figure 5: Hugging Face Deployment — Application Interface with Dual-Mode Inference toggle

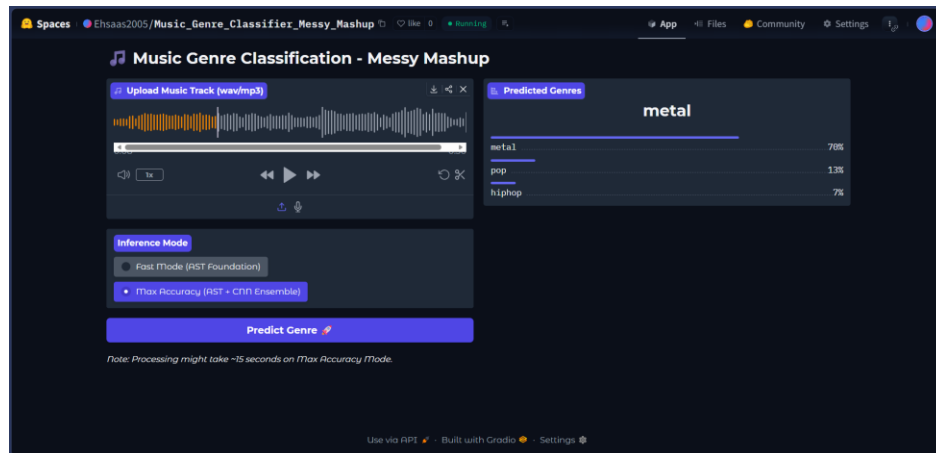


Figure 6: Live Prediction — Metal genre detected with 70% confidence using Max Accuracy mode

9. Conclusion

This project successfully demonstrates that Audio Spectrogram Transformers, when combined with aggressive synthetic data augmentation and multi-model ensembling, can achieve robust genre classification even under extreme audio corruption. The AST backbone alone achieved a validation Macro F1 of 0.9269, decisively outperforming all CNN baselines. The weighted ensemble with 15-pass TTA further stabilized predictions across the full test set. The deployed Hugging Face application provides a practical, secure, and user-friendly interface for real-time music genre classification, validating the complete end-to-end pipeline from training to production deployment.